Maharashtra Education Society's

Institute of Management and Career Courses (IMCC), Pune

AUTONOMOUS

[CGPA-3.32] in Cycle 1

IT13: ADBMS

Practical Assignment 2 Solution Template

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Practical 2: Implementation of different types of functions with suitable examples.

- Number Function
- Aggregate Function
- Character Function
- Conversion Function
- Date Function

LAB PRACTICE ASSIGNMENT:

Create a table EMPLOYEE with the following schema:

(Emp_no, E_name, E_address, E_ph_no, Dept_no, Dept_name,Job_id, Designation, Salary)

Write SQL statements for the following queries.

1. List the E_no, E_name, Salary of all employees working for MANAGER.

SELECT Emp_no, E_name, Salary

FROM EMPLOYEE

WHERE Designation like 'MANAGER';

```
mysql> SELECT Emp_no, E_name, Salary
    -> FROM EMPLOYEE
    -> WHERE Designation like 'MANAGER';
+--------+--------+--------+
| Emp_no | E_name | Salary |
+--------+--------+--------+
|    103 | ALLEN  |  25000 |
+--------+--------+--------+
1 row in set (0.010 sec)
```

2. Display all the details of the employee whose salary is more than 10000.

```
SELECT *
FROM EMPLOYEE
WHERE Salary > 10000;
```

```
mysql> SELECT *
    -> FROM EMPLOYEE
    -> WHERE Salary > 10000;
+--------+--------+-----------+---------+---------+-----------+--------+-------------+--------+--------------+
| Emp_no | E_name | E_address | E_ph_no | Dept_no | Dept_name | Job_id | Designation | Salary | Joining_date |
+--------+--------+-----------+---------+---------+-----------+--------+-------------+--------+--------------+
|    102 | SCOTT  | Mumbai    |  222222 |      20 | SALES     |      2 | ANALYST     |  15000 | 1981-05-01   |
|    103 | ALLEN  | Delhi     |  333333 |      10 | IT        |      3 | MANAGER     |  25000 | 1981-12-03   |
|    105 | KING   | Bangalore |  555555 |      20 | FINANCE   |      5 | PRESIDENT   |  50000 | 1981-12-17   |
|    106 | HARISH | Chennai   |  666666 |      10 | IT        |      6 | ANALYST     |  12000 | 1983-04-10   |
+--------+--------+-----------+---------+---------+-----------+--------+-------------+--------+--------------+
4 rows in set (0.001 sec)
```

3. List the employees in ascending order of Designations of those who joined after 1981.

```
SELECT *
FROM EMPLOYEE
WHERE EXTRACT(YEAR FROM Joining_date) > 1981
ORDER BY Designation;
```

```
mysql> SELECT *
    -> FROM EMPLOYEE
    -> WHERE EXTRACT(YEAR FROM Joining_date) > 1981
    -> ORDER BY Designation;
+--------+--------+-----------+---------+---------+-----------+--------+-------------+--------+--------------+
| Emp_no | E_name | E_address | E_ph_no | Dept_no | Dept_name | Job_id | Designation | Salary | Joining_date |
+--------+--------+-----------+---------+---------+-----------+--------+-------------+--------+--------------+
|    106 | HARISH | Chennai   |  666666 |      10 | IT        |      6 | ANALYST     |  12000 | 1983-04-10   |
|    104 | HARRY  | Pune      |  444444 |      30 | HR        |      4 | CLERK       |   9000 | 1982-06-15   |
+--------+--------+-----------+---------+---------+-----------+--------+-------------+--------+--------------+
2 rows in set (0.003 sec)
```

4. List the employees along with their Experience and Daily Salary.

```
SELECT
E_name,
TIMESTAMPDIFF(YEAR, Joining_date, CURDATE()) AS Experience,
Salary / 30
FROM EMPLOYEE;
```

```
mysql> SELECT
    -> E_name,
    -> TIMESTAMPDIFF(YEAR, Joining_date, CURDATE()) AS Experience,
    -> Salary / 30
    -> FROM EMPLOYEE;
+--------+------------+--------------------+
| E_name | Experience | Salary / 30        |
+--------+------------+--------------------+
| SMITH  |         46 |  266.6666666666667 |
| SCOTT  |         45 |                500 |
| ALLEN  |         44 |  833.3333333333334 |
| HARRY  |         44 |                300 |
| KING   |         44 | 1666.6666666666667 |
| HARISH |         43 |                400 |
+--------+------------+--------------------+
6 rows in set (0.015 sec)
```

5. List the employees who are either 'CLERK' or 'ANALYST '.

```
SELECT *
FROM EMPLOYEE
WHERE Designation IN ('CLERK', 'ANALYST');
```

```
mysql> SELECT *
    -> FROM EMPLOYEE
    -> WHERE Designation IN ('CLERK', 'ANALYST');
+--------+--------+-----------+---------+---------+-----------+--------+-------------+--------+--------------+
| Emp_no | E_name | E_address | E_ph_no | Dept_no | Dept_name | Job_id | Designation | Salary | Joining_date |
+--------+--------+-----------+---------+---------+-----------+--------+-------------+--------+--------------+
|    101 | SMITH  | Pune      |  111111 |      10 | IT        |      1 | CLERK       |   8000 | 1980-01-19   |
|    102 | SCOTT  | Mumbai    |  222222 |      20 | SALES     |      2 | ANALYST     |  15000 | 1981-05-01   |
|    104 | HARRY  | Pune      |  444444 |      30 | HR        |      4 | CLERK       |   9000 | 1982-06-15   |
|    106 | HARISH | Chennai   |  666666 |      10 | IT        |      6 | ANALYST     |  12000 | 1983-04-10   |
+--------+--------+-----------+---------+---------+-----------+--------+-------------+--------+--------------+
4 rows in set (0.009 sec)
```

6. List the employees who joined on 1-MAY-1981, 3-DEC-1981, 17-DEC-1981, 19-JAN-1980.

```
SELECT *
FROM EMPLOYEE
WHERE Joining_date IN ('1981-05-01', '1981-12-03', '1981-12-17', '1980-01-19');
```

```
mysql> SELECT *
    -> FROM EMPLOYEE
    -> WHERE Joining_date IN ('1981-05-01', '1981-12-03', '1981-12-17', '1980-01-19');
+--------+--------+-----------+---------+---------+-----------+--------+-------------+--------+--------------+
| Emp_no | E_name | E_address | E_ph_no | Dept_no | Dept_name | Job_id | Designation | Salary | Joining_date |
+--------+--------+-----------+---------+---------+-----------+--------+-------------+--------+--------------+
|    101 | SMITH  | Pune      |  111111 |      10 | IT        |      1 | CLERK       |   8000 | 1980-01-19   |
|    102 | SCOTT  | Mumbai    |  222222 |      20 | SALES     |      2 | ANALYST     |  15000 | 1981-05-01   |
|    103 | ALLEN  | Delhi     |  333333 |      10 | IT        |      3 | MANAGER     |  25000 | 1981-12-03   |
|    105 | KING   | Bangalore |  555555 |      20 | FINANCE   |      5 | PRESIDENT   |  50000 | 1981-12-17   |
+--------+--------+-----------+---------+---------+-----------+--------+-------------+--------+--------------+
4 rows in set (0.012 sec)
```

7. List the employees who are working for Deptno 10 or20.

```
SELECT *
FROM EMPLOYEE
WHERE Dept_no = 10
OR Dept_no = 20;
```

```
mysql> SELECT *
    -> FROM EMPLOYEE
    -> WHERE Dept_no = 10
    -> OR Dept_no = 20;
+--------+--------+-----------+---------+---------+-----------+--------+-------------+--------+--------------+
| Emp_no | E_name | E_address | E_ph_no | Dept_no | Dept_name | Job_id | Designation | Salary | Joining_date |
+--------+--------+-----------+---------+---------+-----------+--------+-------------+--------+--------------+
|    101 | SMITH  | Pune      |  111111 |      10 | IT        |      1 | CLERK       |   8000 | 1980-01-19   |
|    102 | SCOTT  | Mumbai    |  222222 |      20 | SALES     |      2 | ANALYST     |  15000 | 1981-05-01   |
|    103 | ALLEN  | Delhi     |  333333 |      10 | IT        |      3 | MANAGER     |  25000 | 1981-12-03   |
|    105 | KING   | Bangalore |  555555 |      20 | FINANCE   |      5 | PRESIDENT   |  50000 | 1981-12-17   |
|    106 | HARISH | Chennai   |  666666 |      10 | IT        |      6 | ANALYST     |  12000 | 1983-04-10   |
+--------+--------+-----------+---------+---------+-----------+--------+-------------+--------+--------------+
5 rows in set (0.001 sec)
```

8. List the Enames that start with 'S'.

SELECT E_name

FROM EMPLOYEE

WHERE E_name like 'S%';

```
mysql> SELECT E_name
    -> FROM EMPLOYEE
    -> WHERE E_name like 'S%';
+--------+
| E_name |
+--------+
| SMITH  |
| SCOTT  |
+--------+
2 rows in set (0.000 sec)
```

9. Write a SQL query to find the highest salary. Return the highest salary

SELECT MAX(Salary)

FROM EMPLOYEE;

```
mysql> SELECT MAX(Salary)
    -> FROM EMPLOYEE;
+-------------+
| MAX(Salary) |
+-------------+
|       50000 |
+-------------+
1 row in set (0.003 sec)
```

10. Display the name and the first five characters of the name starting with 'H'. List all the employees except 'PRESIDENT' & 'MGR" in ascending order of Salaries.

```
SELECT E_name, SUBSTRING(E_name, 1, 5)
FROM EMPLOYEE
WHERE E_name like 'H____%'
AND Designation NOT IN ('MGR', 'PRESIDENT')
ORDER BY Salary;
```

```
mysql> SELECT E_name, SUBSTRING(E_name, 1, 5)
    -> FROM EMPLOYEE
    -> WHERE E_name like 'H____%'
    -> AND Designation NOT IN ('MGR', 'PRESIDENT')
    -> ORDER BY Salary;
+--------+-------------------------+
| E_name | SUBSTRING(E_name, 1, 5) |
+--------+-------------------------+
| HARRY  | HARRY                   |
| HARISH | HARIS                   |
+--------+-------------------------+
2 rows in set (0.005 sec)
```

11. Display the minimum and maximum salary and Experience of the employee.

```
SELECT MAX(Salary) AS MAX_SALARY,
MIN(Salary) AS MIN_SALARY,
MAX(TIMESTAMPDIFF(YEAR, Joining_date, CURDATE())) AS MAX_EXPERIANCE,
MIN(TIMESTAMPDIFF(YEAR, Joining_date, CURDATE())) AS MIN_EXPERIANCE
FROM EMPLOYEE;
```

```
mysql> SELECT MAX(Salary) AS MAX_SALARY,
    -> MIN(Salary) AS MIN_SALARY,
    -> MAX(TIMESTAMPDIFF(YEAR, Joining_date, CURDATE())) AS MAX_EXPERIANCE,
    -> MIN(TIMESTAMPDIFF(YEAR, Joining_date, CURDATE())) AS MIN_EXPERIANCE
    -> FROM EMPLOYEE;
+------------+------------+----------------+----------------+
| MAX_SALARY | MIN_SALARY | MAX_EXPERIANCE | MIN_EXPERIANCE |
+------------+------------+----------------+----------------+
|      50000 |       8000 |             46 |             43 |
+------------+------------+----------------+----------------+
1 row in set (0.009 sec)
```

12. Display the total salary of all employees.

SELECT SUM(Salary)

FROM EMPLOYEE;

```
mysql> SELECT SUM(Salary)
    -> FROM EMPLOYEE;
+-------------+
| SUM(Salary) |
+-------------+
|      119000 |
+-------------+
1 row in set (0.000 sec)
```

13. Write a query to display the average salary of employees working in IT department.

SELECT AVG(Salary)

FROM EMPLOYEE

WHERE Dept_name = 'IT';

```
mysql> SELECT AVG(Salary)
    -> FROM EMPLOYEE
    -> WHERE Dept_name = 'IT';
+-------------+
| AVG(Salary) |
+-------------+
|       15000 |
+-------------+
1 row in set (0.010 sec)
```

14. Display the Distinct salary of an employee.

SELECT DISTINCT Salary

FROM EMPLOYEE;

```
mysql> SELECT DISTINCT Salary
    -> FROM EMPLOYEE;
+--------+
| Salary |
+--------+
|   8000 |
|  15000 |
|  25000 |
|   9000 |
|  50000 |
|  12000 |
+--------+
6 rows in set (0.031 sec)
```

15. Drop the database.

DROP DATABASE ASS_2;

```
mysql>
mysql> DROP DATABASE ASS_2;
Query OK, 1 row affected (0.034 sec)
```